DEPARTMENT OF MATHEMATICS AND APPLIED MATHEMATICS
UNIVERSITY OF CAPE TOWN

Mathematics III **Complex Analysis [3CA]** **2017**

Class Test 1

Date : 20 September 2017 Marks available: 51
Time : 6:00 – 7:30 pm Full marks: 50

No calculators may be used.

1. Read the following theorem and the outline of its proof:

Theorem. *Suppose that a complex-valued function f is analytic on a region $R \subseteq \mathbb{C}$. If* $\operatorname{Im}(f(z))$ *is constant for all $z \in R$, then $f(z)$ is constant on R.*

Proof. Suppose that $f(z) = u(x, y) + iv(x, y)$, where $z = x + iy$ and x, y, u, v are all real. In class we showed that

$$f'(z) = \frac{\partial u}{\partial x} + i\frac{\partial v}{\partial x} = \frac{\partial v}{\partial y} - i\frac{\partial u}{\partial y}. \qquad (*)$$

Since $v(x, y)$ is constant on R, we have that $\frac{\partial v}{\partial x} = \frac{\partial v}{\partial y} = 0$ throughout R, and hence $\frac{\partial u}{\partial x} = \frac{\partial u}{\partial y} = 0$ at every point $z \in R$. Consider now an open disc D centred at $z_0 \in R$ and contained in R. Given any point $z \in D$, we can use the Mean Value Theorem to prove that $f(z) = f(z_0)$. Thus f is constant on D. Put $G = \{z \in R : f(z) = f(z_0)\}$. Notice that G is an open set in $\mathbb{C}$. However, $R \setminus G = \{z \in R : f(z) \neq f(z_0)\}$ is also open. This is only possible if $R \setminus G = \emptyset$, so $G = R$.

Now answer the following questions:

(a) Define what is meant by a *region* in $\mathbb{C}$.

(b) What do we mean when we say that f is *analytic* on R?

(c) Derive the Cauchy–Riemann equations $(*)$.

(d) State the Mean Value Theorem. Does the MVT hold for complex-valued functions? (Either prove that it does, or use an example to show that it doesn't.)

(e) Explain carefully how the MVT can be used to prove that $f(z) = f(z_0)$ for all z in the open disc D.

(f) Why is $G = \{z \in R : f(z) = f(z_0)\}$ an open set in $\mathbb{C}$?

(g) Why is $R \setminus G = \{z \in R : f(z) \neq f(z_0)\}$ open?

(h) Explain why this is only possible if $R \setminus G = \emptyset$. [2, 1, 5, 4, 6, 4, 2, 2]

2. Find the radius of convergence of each of the following power series.

(a) $\displaystyle\sum_{k=1}^{\infty} \frac{k!}{k^k}(z-1)^k$ (b) $\displaystyle\sum_{k=0}^{\infty} 3^k z^{3^k}$

[4, 3]

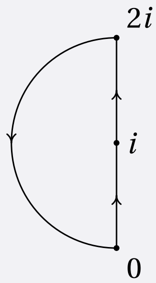
3. Write down a power series of the form $\sum_{k=0}^{\infty} a_k z^k$ for

   (a) $\dfrac{1-iz}{1+iz}$ (b) $\dfrac{1}{1+3z+9z^2}$

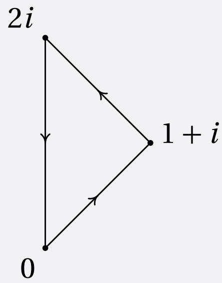
   In each case, give the domain of convergence. (Give reasons for your answer.) [4, 5]

4. Calculate each of the following integrals using the illustrated path:

   (a) $\displaystyle\int_\gamma \overline{z}\,\mathrm{d}z$



   (b) $\displaystyle\int_\gamma \frac{\tan z}{1-z^2}\,\mathrm{d}z$



[9]

Class Test 1 solutions

1.(a) A *region* is a non-empty open connected subset of $\mathbb{C}$.

(b) f is *analytic* on a region R if it is differentiable at each point in R.

(c) Since f is analytic on R,

$$f'(z) = \lim_{h\to 0} \frac{f(z+h) - f(z)}{h}$$

exists at each point $z \in R$. It follows that

$$\begin{aligned}
f'(z) &= \lim_{\substack{h\to 0\\ h\in\mathbb{R}}} \frac{f(z+h)-f(z)}{h}\\
&= \lim_{\substack{h\to 0\\ h\in\mathbb{R}}} \frac{u(x+h,y)-u(x,y)}{h} + i\lim_{\substack{h\to 0\\ h\in\mathbb{R}}} \frac{v(x+h,y)-v(x,y)}{h}\\
&= \frac{\partial u}{\partial x} + i\frac{\partial v}{\partial y}.
\end{aligned}$$

Similarly,

$$\begin{aligned}
f'(z) &= \lim_{\substack{k\to 0\\ k\in\mathbb{R}}} \frac{f(z+ik)-f(z)}{ik}\\
&= -i\lim_{\substack{k\to 0\\ k\in\mathbb{R}}} \frac{f(z+ik)-f(z)}{k}\\
&= -i\lim_{\substack{k\to 0\\ k\in\mathbb{R}}} \frac{u(x,y+k)-u(x,y)}{k} + i\lim_{\substack{k\to 0\\ k\in\mathbb{R}}} \frac{v(x,y+k)-v(x,y)}{h}\\
&= -i\frac{\partial u}{\partial x} + \frac{\partial v}{\partial y}.
\end{aligned}$$

(d) The Mean Value Theorem says that if a real-valued function is differentiable at each point in an open interval $I \subseteq \mathbb{R}$, then for any two points $a, b \in I$, there is a point c between a, b with

$$f'(c) = \frac{f(b)-f(a)}{b-a}.$$

This theorem does not work for complex-valued functions. Consider the function $f(z) = z^4$. We have that $\dfrac{f(i)-f(1)}{i-1} = 0$, but the only point with $f'(z) = 4z^3 = 0$ is $z = 0$, which is not between 1 and i.

(e) Let $z_0 = x_0 + iy_0$ and $z = x + iy$, where $x_0, y_0, x, y \in \mathbb{R}$. Since D is an open disc, $g(t) = u(t, y_0)$ is a real-valued function defined on an open interval containing x_0 and x. The fact that $\dfrac{\partial u}{\partial x} = 0$ means that $g'(t) = 0$ for all t between x_0 and x. Thus the MVT says that

$$u(x, y_0) = g(x) = g(x_0) = u(x_0, y_0).$$

Similarly, we can define a real-valued function on an open interval containing y_0 and y by putting $h(t) = u(x,t)$. Since $\dfrac{\partial u}{\partial v} = 0$ on D, we get that $h'(t) = 0$ for all t between y_0 and y. Thus

$$x(x,y) = h(y) = h(y_0) = u(x,y_0) = u(x_0,y_0).$$

Since $\mathrm{Im}(f(z))$ is constant on R, we already know that $v(x,y) = v(x_0,y_0)$. Thus $f(z) = f(z_0)$ for all $z \in D$.

(f) Let z_1 be any point in G. Since $z_1 \in R$ and R is open, there is an open disc centred at z_1 contained in R. The first part of the proof shows that any point z in this disc must satisfy $f(z) = f(z_1)$. But then $f(z) = f(z_1) = f(z_0)$, so $z \in G$. Thus every point in G is the centre of a disc contained in G. This shows G is open.

(g) It is possible to prove this using the same argument as in (f). However, it is also possible to use the fact that $R \setminus G$ is the pre-image of the open set $\{w \in \mathbb{C} : w \neq f(z_0)\}$.

(h) G and $R \setminus G$ are disjoint open subsets of the connected set R. $G \neq \emptyset$ since $z_0 \in G$. Thus $R \setminus G = \emptyset$.

2.(a) By the ratio formula,

$$R = \lim_{k\to\infty} \frac{k!}{k^k} \cdot \frac{(k+1)^{k+1}}{(k+1)!} = \lim_{k\to\infty} \left(\frac{k+1}{k}\right)^k = \lim_{k\to\infty} \left(1 + \frac{1}{k}\right)^k = e.$$

(b) Notice that $a_n = n$ if $n = 3^k$ for some k; otherwise $a_n = 0$. The Cauchy-Hadamard formula says that

$$\frac{1}{R} = \limsup_n n^{1/n} = 1.$$

Thus $R = 1$.

3. We shall use the formula for a geometric progression:

$$\frac{1}{1-w} = 1 + w + w^2 + w^3 + \cdots = \sum_{k=0}^{\infty} w^k.$$

This converges if and only if $|w| < 1$.

(a) There are several ways of tackling this one.

$$\begin{aligned}
\frac{1-iz}{1+iz} &= (1-iz) \cdot \frac{1}{1+iz} \\
&= (1-iz) \sum_{k=0}^{\infty} (-iz)^k \qquad \text{(putting } w = -iz\text{)} \\
&= \sum_{k=0}^{\infty} (-iz)^k - iz \sum_{k=0}^{\infty} (-iz)^k \\
&= \sum_{k=0}^{\infty} (-iz)^k + \sum_{k=0}^{\infty} (-iz)^{k+1} \\
&= 1 + \sum_{k=1}^{\infty} (-i)^k z^k + \sum_{k=1}^{\infty} (-i)^k z^k \\
&= 1 + \sum_{k=1}^{\infty} 2(-i)^k z^k.
\end{aligned}$$

(You can simplify these coefficients if you wish.)

By the above, this converges $\iff |w| < 1 \iff |z| < 1$.

(b) If $z \neq \frac{1}{3}$, then $\dfrac{1}{1+3z+9z^2} = \dfrac{1-3z}{1-27z^3}$. Thus, using $w = 27z^3 = (3z)^3$,

$$\frac{1}{1+3z+9z^2} = (1-3z)\sum_{k=0}^{\infty}(3z)^{3k} = \sum_{k=0}^{\infty}(3z)^{3k} - \sum_{k=0}^{\infty}(3z)^{3k+1} = \sum_{k=0}^{\infty} a_k z^k,$$

where

$$a_k = \begin{cases} 3^k & \text{if } k \equiv 0 \mod 3, \\ -3^k & \text{if } k \equiv 1 \mod 3, \\ 0 & \text{if } k \equiv 2 \mod 3. \end{cases}$$

The series converges $\iff |w| < 1 \iff \left|(3z)^3\right| < 1 \iff |z| < \frac{1}{3}$.

4.(a) We split the path into pieces. The straight piece can be parametrised by $\gamma_1(t) = it$, where $0 \leqslant t \leqslant 2$, so

$$\int_{\gamma_1} \overline{z}\, \mathrm{d}z = \int_0^2 (-it)\, i\, \mathrm{d}t = 2.$$

The semicircle can be parametrised by $\gamma_2(t) = i + e^{it}$, where $\frac{\pi}{2} \leqslant t \leqslant \frac{3\pi}{2}$. This yields

$$\begin{aligned}\int_{\gamma_2} \overline{z}\, \mathrm{d}z &= \int_{\pi/2}^{3\pi/2} \left(-i + e^{-it}\right) ie^{it} \mathrm{d}t \\ &= \int_{\pi/2}^{3\pi/2} e^{it}\, \mathrm{d}t + \int_{\pi/2}^{3\pi/2} i\, \mathrm{d}t \\ &= -2 + \pi i.\end{aligned}$$

Thus $\displaystyle\int_{\gamma} \overline{z}\, \mathrm{d}z = \pi i$.

(b) The integrand is analytic inside and on the curve. The Cauchy-Goursat Theorem says that the integral is 0.